

Autonomous Vehicle Simulation (AVS) Laboratory, University of Colorado

Basilisk Technical Memorandum

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GUIDANCE MODULE FOR CELESTIAL TWO BODY POINT

Prepared by	M. Cols

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Scope/Contents
This module point one body-fixed axis towards a primary celestial object. The secondary goal is to point a second body-fixed axis towards another celestial object. For example, the goal is to point the sensor towards the center of a planet while doing the best to keep the solar panel normal point at the sun.

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Contents

1 Model Description **1**
1.1 Module Goal . . . 1
1.2 Special Case: No Secondary Constraint Applicable . . . 2
1.3 Reference Frame Definition . . . 3
1.4 Base Vectors Time Derivatives . . . 3
1.5 Angular Velocity and Acceleration Descriptions . . . 4

2 Module Functions **5**

3 Module Assumptions and Limitations **5**

4 Test Description and Success Criteria **5**
4.1 Test 1 . . . 5
4.2 Test 2 . . . 5
4.3 Finite-Difference Regression Tests . . . 6

5 Test Parameters **6**

6 Test Results **6**

7 User Guide **6**
7.1 Input/Output Messages . . . 6
7.2 Module Parameters and States . . . 7

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1 Model Description

1.1 Module Goal

This module computes a reference whose aim is to track the center of a primary target, e.g. pointing the communication antenna at the Earth, at the same time of trying to meet a secondary constraint as best as possible, e.g. a solar panel normal axis pointing the closest in the direction of the Sun. It is important to note that two pointing conditions in a three-dimensional space compose an overdetermined problem. Thus, the main constraint is always priorized over the secondary one so the former can always be met.

Figure 1 shows the case where Mars is the primary celestial body and the Sun is the secondary one. Note that the origin of the desired reference frame $\mathcal{R}$ is the position of the spacecraft.

Assuming knowledge of the position of the spacecraft $\boldsymbol{r}_{B/N}$ and the involved celestial bodies, $\boldsymbol{R}_{P1}$ and $\boldsymbol{R}_{P2}$ (all of them relative to the inertial frame $\mathcal{N}$ and expressed in inertial frame components), the relative position of the celestial bodies with respect to the spacecraft is obtained by simple subtraction:

$$\boldsymbol{R}_{P1} = \boldsymbol{R}_P - \boldsymbol{r}_{B/N} \tag{1a}$$

$$\boldsymbol{R}_{P2} = \boldsymbol{R}_S - \boldsymbol{r}_{B/N} \tag{1b}$$

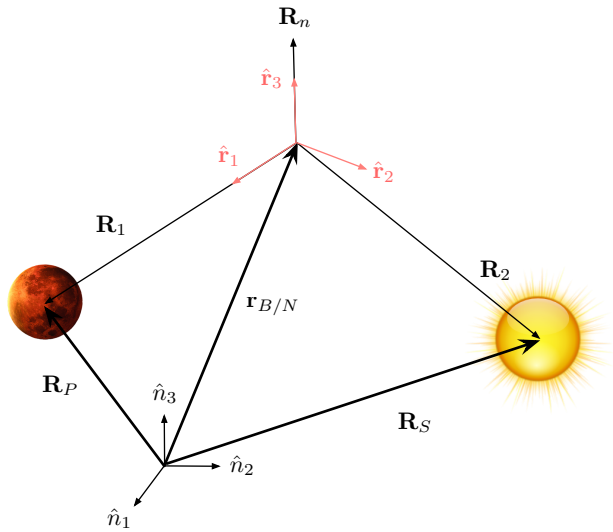


Fig. 1: Illustration of the restricted two-body pointing reference frame $\mathcal{R} : \{\hat{\boldsymbol{r}}_1, \hat{\boldsymbol{r}}_1, \hat{\boldsymbol{r}}_2\}$ and the inertial frame $\mathcal{N} : \{\hat{\boldsymbol{n}}_1, \hat{\boldsymbol{n}}_1, \hat{\boldsymbol{n}}_2\}$.

In analogy, the inertial derivatives of these position vectors are obtained:

$$\boldsymbol{v}_{P1} = \boldsymbol{v}_P - \boldsymbol{v}_{B/N} \tag{2a}$$
$$\boldsymbol{v}_{P2} = \boldsymbol{v}_S - \boldsymbol{v}_{B/N} \tag{2b}$$
$$\boldsymbol{a}_{P1} = \boldsymbol{a}_P - \boldsymbol{a}_{B/N} \tag{2c}$$
$$\boldsymbol{a}_{P2} = \boldsymbol{a}_S - \boldsymbol{a}_{B/N} \tag{2d}$$

The normal vector $\boldsymbol{R}_n$ of the plane defined by $\boldsymbol{R}_{P1}$ and $\boldsymbol{R}_{P2}$ is computed through:

$$\boldsymbol{R}_n = \boldsymbol{R}_{P1} \times \boldsymbol{R}_{P2} \tag{3}$$

The inertial time derivative of $\boldsymbol{R}_n$ is found using the chain differentiation rule:

$$\boldsymbol{v}_n = \boldsymbol{v}_{P1} \times \boldsymbol{R}_{P2} + \boldsymbol{R}_{P1} \times \boldsymbol{v}_{P2} \tag{4}$$

And the second time derivative:

$$\boldsymbol{a}_n = \boldsymbol{a}_{P1} \times \boldsymbol{R}_{P2} + \boldsymbol{R}_{P1} \times \boldsymbol{a}_{P2} + 2\boldsymbol{v}_{P1} \times \boldsymbol{v}_{P2} \tag{5}$$

1.2 Special Case: No Secondary Constraint Applicable

If there is no incoming message with a secondary celestial body pointing condition or if the constrain is not valid, an artificial three-dimensional frame is defined instead. Note that a single condition pointing leaves one degree of freedom, hence standing for an underdetermined attitude problem. A secondary constrain is considered to be invalid if the angle between $\boldsymbol{R}_{P1}$ and $\boldsymbol{R}_{P2}$ is, in absolute value, minor than a set threshold. This could be the case where the primary and secondary celestial bodies are aligned as seen by the spacecraft. In such situation, the primary pointing axis would already satisfy both the primary and the secondary constraints.

Since the main algorithm of this module, which is developed in the following sections, assumes two conditions, the second one is arbitrarily set as following:

$$\boldsymbol{R}_{P2} = \boldsymbol{R}_{P1} \times \boldsymbol{v}_{P1} \equiv \boldsymbol{h}_{P1} \tag{6}$$

By setting the secondary constrain to have the direction of the angular momentum vector $\boldsymbol{h}_{P1}$, it is assured that it will always be valid ($\boldsymbol{R}_{P1}$ and $\boldsymbol{R}_{P2}$ are now perpendicular). The first and second time derivatives are steadily computed:

$$\boldsymbol{v}_{P2} = \boldsymbol{R}_{P1} \times \boldsymbol{a}_{P1} \tag{7}$$

$$\boldsymbol{a}_{P2} = \boldsymbol{v}_{P1} \times \boldsymbol{a}_{P1} \tag{8}$$

1.3 Reference Frame Definition

As illustrated in Figure 1, the base vectors of the desired reference frame $\mathcal{R}$ are defined as following:

$$\hat{\boldsymbol{r}}_1 = \frac{\boldsymbol{R}_{P1}}{|\boldsymbol{R}_{P1}|} \tag{9a}$$

$$\hat{\boldsymbol{r}}_3 = \frac{\boldsymbol{R}_n}{|\boldsymbol{R}_n|} \tag{9b}$$

$$\hat{\boldsymbol{r}}_2 = \hat{\boldsymbol{r}}_3 \times \hat{\boldsymbol{r}}_1 \tag{9c}$$

Since the position vectors are given in terms of inertial $\mathcal{N}$-frame components, the DCM from the inertial frame $\mathcal{N}$ to the desired reference frame $\mathcal{R}$ is:

$$[RN] = \begin{bmatrix} {}^{\mathcal{N}}\hat{\boldsymbol{r}}_1^T \\ {}^{\mathcal{N}}\hat{\boldsymbol{r}}_2^T \\ {}^{\mathcal{N}}\hat{\boldsymbol{r}}_3^T \end{bmatrix} \tag{10}$$

1.4 Base Vectors Time Derivatives

Dots denote inertial time derivatives. In the matrix expressions below, all vectors and their derivatives are expressed in inertial $\mathcal{N}$-frame components. Thus, the arrays `r1_hat_N`, `dr1_hat_N`, and `ddr1_hat_N` represent a basis vector and its first two inertial derivatives, all in the same component frame.

Define the projection matrices

$$C_i = I_{3\times 3} - \hat{\boldsymbol{r}}_i \hat{\boldsymbol{r}}_i^T, \qquad i \in \{1, 3\}.$$

These are projections orthogonal to the indicated basis vector, not direction cosine matrices. The first derivatives are

$$\dot{\hat{\boldsymbol{r}}}_1 = C_1 \frac{\boldsymbol{v}_{P1}}{|\boldsymbol{R}_{P1}|}, \tag{11a}$$

$$\dot{\hat{\boldsymbol{r}}}_3 = C_3 \frac{\boldsymbol{v}_n}{|\boldsymbol{R}_n|}, \tag{11b}$$

$$\dot{\hat{\boldsymbol{r}}}_2 = \dot{\hat{\boldsymbol{r}}}_3 \times \hat{\boldsymbol{r}}_1 + \hat{\boldsymbol{r}}_3 \times \dot{\hat{\boldsymbol{r}}}_1. \tag{11c}$$

For example, differentiating the normalized primary direction gives

$$\dot{\hat{\boldsymbol{r}}}_1 = \frac{d}{dt}\left(\frac{\boldsymbol{R}_{P1}}{|\boldsymbol{R}_{P1}|}\right), \tag{12a}$$

$$= \frac{\boldsymbol{v}_{P1}}{|\boldsymbol{R}_{P1}|} - \frac{\boldsymbol{R}_{P1}(\boldsymbol{R}_{P1} \cdot \boldsymbol{v}_{P1})}{|\boldsymbol{R}_{P1}|^3}, \tag{12b}$$

$$= \frac{1}{|\boldsymbol{R}_{P1}|}(I_{3\times 3} - \hat{\boldsymbol{r}}_1 \hat{\boldsymbol{r}}_1^T)\boldsymbol{v}_{P1}. \tag{12c}$$

The second derivatives are

$$\ddot{\hat{\boldsymbol{r}}}_1 = \frac{C_1\boldsymbol{a}_{P1} - (2\dot{\hat{\boldsymbol{r}}}_1\hat{\boldsymbol{r}}_1^T + \hat{\boldsymbol{r}}_1\dot{\hat{\boldsymbol{r}}}_1^T)\boldsymbol{v}_{P1}}{|\boldsymbol{R}_{P1}|}, \tag{13a}$$

$$\ddot{\hat{\boldsymbol{r}}}_3 = \frac{C_3\boldsymbol{a}_n - (2\dot{\hat{\boldsymbol{r}}}_3\hat{\boldsymbol{r}}_3^T + \hat{\boldsymbol{r}}_3\dot{\hat{\boldsymbol{r}}}_3^T)\boldsymbol{v}_n}{|\boldsymbol{R}_n|}, \tag{13b}$$

$$\ddot{\hat{\boldsymbol{r}}}_2 = \ddot{\hat{\boldsymbol{r}}}_3 \times \hat{\boldsymbol{r}}_1 + \hat{\boldsymbol{r}}_3 \times \ddot{\hat{\boldsymbol{r}}}_1 + 2\dot{\hat{\boldsymbol{r}}}_3 \times \dot{\hat{\boldsymbol{r}}}_1. \tag{13c}$$

In particular, the two outer-product terms inside each parenthesis have the same sign. This follows from

$$\dot{C}_1 = -\dot{\hat{\boldsymbol{r}}}_1\hat{\boldsymbol{r}}_1^T - \hat{\boldsymbol{r}}_1\dot{\hat{\boldsymbol{r}}}_1^T, \tag{14a}$$

$$\frac{d}{dt}|\boldsymbol{R}_{P1}| = \hat{\boldsymbol{r}}_1 \cdot \boldsymbol{v}_{P1}, \tag{14b}$$

$$\ddot{\hat{\boldsymbol{r}}}_1 = \frac{C_1\boldsymbol{a}_{P1} + \dot{C}_1\boldsymbol{v}_{P1} - \dot{\hat{\boldsymbol{r}}}_1(\hat{\boldsymbol{r}}_1 \cdot \boldsymbol{v}_{P1})}{|\boldsymbol{R}_{P1}|}. \tag{14c}$$

1.5 Angular Velocity and Acceleration Descriptions

The angular velocity projections onto the reference basis satisfy

$$\boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_1 = \hat{\boldsymbol{r}}_3 \cdot \dot{\hat{\boldsymbol{r}}}_2, \tag{15a}$$

$$\boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_2 = \hat{\boldsymbol{r}}_1 \cdot \dot{\hat{\boldsymbol{r}}}_3, \tag{15b}$$

$$\boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_3 = \hat{\boldsymbol{r}}_2 \cdot \dot{\hat{\boldsymbol{r}}}_1. \tag{15c}$$

Differentiating the left-hand sides introduces the terms $\boldsymbol{\omega}_{R/N} \cdot \dot{\hat{\boldsymbol{r}}}_i$. Each term is zero because the basis rotates with the reference frame:

$$\dot{\hat{\boldsymbol{r}}}_i = \boldsymbol{\omega}_{R/N} \times \hat{\boldsymbol{r}}_i, \qquad \boldsymbol{\omega}_{R/N} \cdot \dot{\hat{\boldsymbol{r}}}_i = 0.$$

Consequently, the angular acceleration projections are

$$\dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_1 = \dot{\hat{\boldsymbol{r}}}_3 \cdot \dot{\hat{\boldsymbol{r}}}_2 + \hat{\boldsymbol{r}}_3 \cdot \ddot{\hat{\boldsymbol{r}}}_2, \tag{16a}$$

$$\dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_2 = \dot{\hat{\boldsymbol{r}}}_1 \cdot \dot{\hat{\boldsymbol{r}}}_3 + \hat{\boldsymbol{r}}_1 \cdot \ddot{\hat{\boldsymbol{r}}}_3, \tag{16b}$$

$$\dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_3 = \dot{\hat{\boldsymbol{r}}}_2 \cdot \dot{\hat{\boldsymbol{r}}}_1 + \hat{\boldsymbol{r}}_2 \cdot \ddot{\hat{\boldsymbol{r}}}_1. \tag{16c}$$

These scalar projections are the R-frame components, although the implementation evaluates each dot product using basis vectors and derivatives in N-frame components:

$${}^{\mathcal{R}}\boldsymbol{\omega}_{R/N} = \begin{bmatrix} \boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_1 \\ \boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_2 \\ \boldsymbol{\omega}_{R/N} \cdot \hat{\boldsymbol{r}}_3 \end{bmatrix}. \tag{17}$$

Similarly,

$${}^{\mathcal{R}}\dot{\boldsymbol{\omega}}_{R/N} = \begin{bmatrix} \dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_1 \\ \dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_2 \\ \dot{\boldsymbol{\omega}}_{R/N} \cdot \hat{\boldsymbol{r}}_3 \end{bmatrix}. \tag{18}$$

Dotting `omega_RN_R` with `dr1_hat_N`, or either of the other basis derivatives, mixes component frames and does not evaluate the zero-valued geometric term. The implementation omits these terms.

Finally, the output message contains N-frame components. Since $[RN]$ maps N to R, the inverse transformation is $[NR] = [RN]^T$:

$$^{\mathcal{N}}\boldsymbol{\omega}_{R/N} = [RN]^T\, {}^{\mathcal{R}}\boldsymbol{\omega}_{R/N}, \tag{19a}$$

$$^{\mathcal{N}}\dot{\boldsymbol{\omega}}_{R/N} = [RN]^T\, {}^{\mathcal{R}}\dot{\boldsymbol{\omega}}_{R/N}. \tag{19b}$$

2 Module Functions

- **parseInputMessages**: This method takes the navigation translational info as well as the spice data of the primary celestial body and, if applicable, the second one, and computes the relative state vectors necessary to create the restricted 2-body pointing reference frame.
- **computeCelestialTwoBodyPoint**: This method takes the spacecraft and points a specified axis at a named celestial body specified in the configuration data. It generates the commanded attitude and assumes that the control errors are computed downstream.

3 Module Assumptions and Limitations

The module sets both celestial-body accelerations relative to the spacecraft to zero. Its reference rate and acceleration describe the instantaneous motion under that assumption; the input messages do not supply relative accelerations.

The primary relative position must be nonzero. If the secondary direction is absent or within `singularityThresh` of parallel or antiparallel to the primary direction, the fallback uses the primary relative position crossed with its relative velocity. This cross product must be nonzero, so purely radial relative motion does not define the fallback frame. Straight-line trajectories are otherwise supported when the required directions are well defined.

4 Test Description and Success Criteria

The mathematics in this module are straight forward and can be tested in a series of input and output evaluation tests.

4.1 Test 1

The first test does a single-body celestial point. It places a spacecraft around Earth on an orbit with parameters given in 2. The earth position and velocity vectors are both set to the zero vectors.

Table 2: Spacecraft Orbital Parameters

Orbital Parameter	Value
a	$2.8R_{\text{earth}}$
e	0
i	0°
Ω	0°
ω	0°
f	60°

4.2 Test 2

The second test does the 2-body celestial point. It uses the same parameters at above and sets a second planet with zero velocity and position vector:

$$^{\mathcal{N}}\boldsymbol{r}_2 = \begin{bmatrix} 500 & 500 & 500 \end{bmatrix}^T \text{(m)} \tag{20}$$

4.3 Finite-Difference Regression Tests

Eight additional cases check the derivatives independently of the analytic truth model. The primary body starts at $[7, 1, -2]^T\ 10^6$ m with velocity $[-1, 7, 2]^T\ 10^3$ m/s. The moving secondary body starts at $[-3, 8, 4]^T\ 10^6$ m with velocity $[5, 1, -3]^T\ 10^3$ m/s. The spacecraft is at rest at the origin. Both bodies follow straight-line trajectories, consistent with the module's zero-relative-acceleration assumption.

The cases cover the moving secondary body, an absent secondary message, and secondary directions that remain parallel or antiparallel to the primary direction. Each configuration is evaluated at 0 s and 250 s, away from any fallback switching boundary. A central-difference step of 0.1 s is used.

The rate check differentiates the basis obtained from the output attitude and uses

$$^{\mathcal{N}}\boldsymbol{\omega}_{R/N} = \frac{1}{2}\sum_{i=1}^{3} {}^{\mathcal{N}}\hat{\boldsymbol{r}}_i \times {}^{\mathcal{N}}\dot{\hat{\boldsymbol{r}}}_i.$$

The acceleration check differentiates the published N-frame angular rate. Absolute tolerances are 10^{-11} rad/s for rate and 10^{-13} rad/s^2 for acceleration. These cases detect the component-frame mismatch from issue 1618 without copying the module's angular-acceleration equations.

5 Test Parameters

For the first two tests, the tested parameters are listed in Table 3.

Table 3: Error tolerance for each test.

Output Value Tested	Tolerated Error
$\boldsymbol{\sigma}_{R/N}$	1e-10
$^{\mathcal{N}}\boldsymbol{\omega}_{R/N}$	1e-10
$^{\mathcal{N}}\dot{\boldsymbol{\omega}}_{R/N}$	1e-10

6 Test Results

The first two tests generate the following results. The finite-difference cases are checked by pytest:

Table 4: Test results

Check	Pass/Fail
1.1	PASSED
1.2	PASSED
1.3	PASSED
2.1	PASSED
2.2	PASSED
2.3	PASSED

7 User Guide

7.1 Input/Output Messages

The module has 2 required input messages, 1 optional input message and 1 output message:

- `inputNavDataName` – This input message, of type `NavTransIntMsg`, provide the translational-navigation states for the spacecraft.
- `inputCelMessName` – This input message, of type `EphemerisIntMsg`, receives the first planet states for pointing

- `inputSecMessName` – (Optional) This input message, of type `EphemerisIntMsg`, receives the second planet states for pointing
- `outputDataName` – This output message, of type `AttRefFswMsg`, writes out the attitude, rate, and inertial derivative of the rate in order to perform control.

7.2 Module Parameters and States

Outside of the message names, this module only has one other parameter:

- `singularityThresh` - This parameter determines the threshold after which two vectors are considered collinear.